

KIDS' PLAYROOM BUILD

Corner Loft · Ranger's Outpost Playhouse · bridge-ready

CONSTRUCTION DOCUMENT SET · dimensional lumber + stock hardware · generated from the parametric Blender model

ROOM 13'-6" x 13'-5" CEILING 8'-0" PLATFORM 3'-8" GUARD 3'-0" GUARD OPENINGS 2.59"-2.70"

DESIGNED FOR 18 MONTHS THROUGH ABOUT 8 YEARS — IN THREE PHASES

READ FIRST — FIVE THINGS IN THE BRIEF THAT DO NOT WORK AS WRITTEN

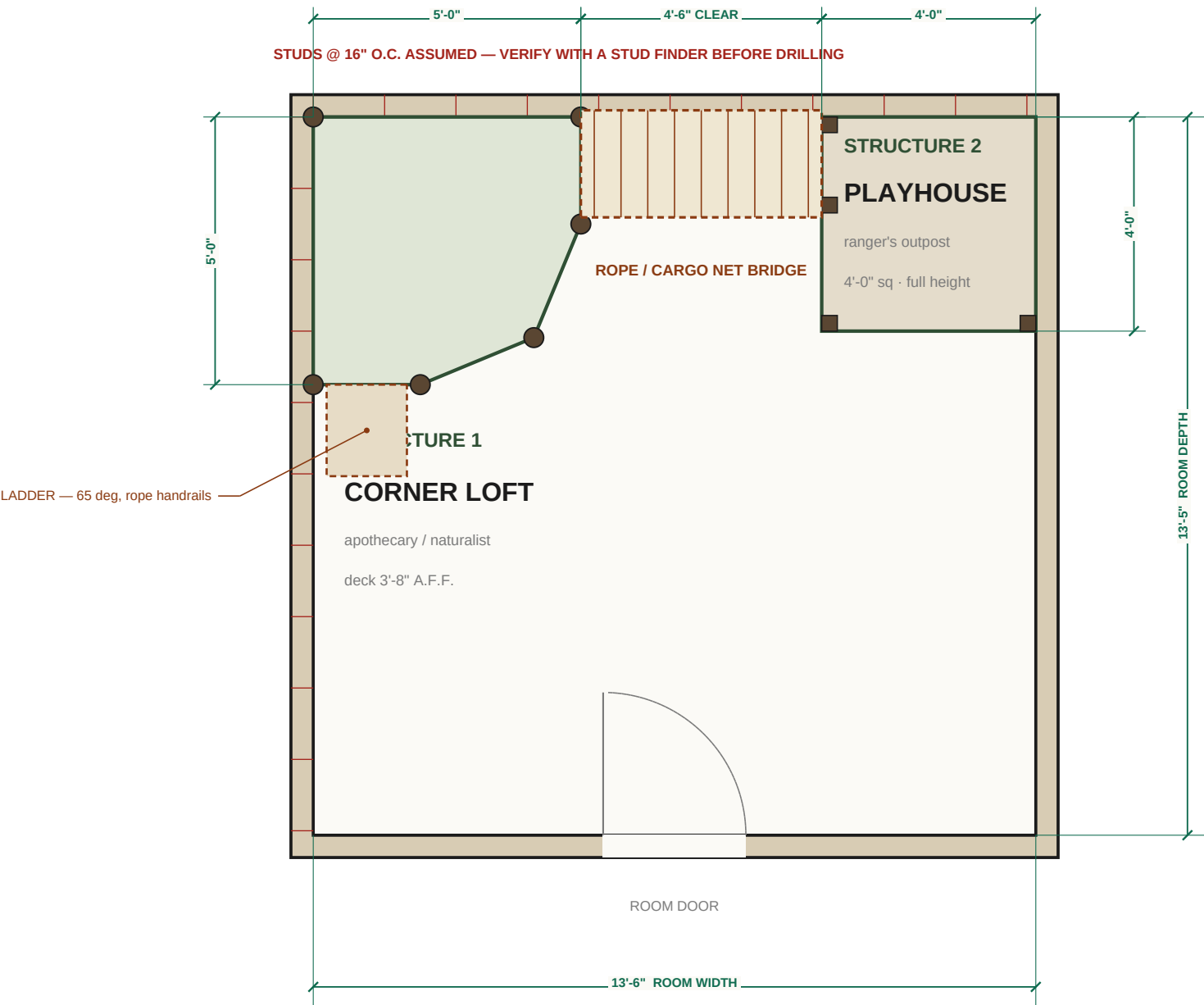
1. This has to work for an 18-month-old AND a four-year-old. That drove a redesign. BRIEF one structure, one age CPSC caps a TODDLER platform (6-23 months) at 32". This deck is 44". You cannot make a 44" loft safe for an 18-month-old, so the loft is not his. The enclosed ground den and the playhouse are, from day one. The ladder LIFTS OFF, which is what actually keeps him out of the loft. Full phase table on this sheet.	4. The knee wall studs are 34-3/4", not 40-42" BRIEF studs ~40-42" tall Same arithmetic. The wall dies into the underside of the joists at 37-3/4", and two plates eat 3" of that. Cut one, hold it in place, and confirm before you cut the other fifteen.
2. The rope net guardrail had to go. It is a head-entrapment hazard. BRIEF rope net railing, 4-6" mesh CPSC: any opening between 3.5" and 9" lets a small body through and catches the head. A 4-6" mesh fails both probes. Vertical balusters at under 3-3/8" clear instead, with a 2x4 toe board closing the gap at deck level. Rope stays as tight decorative wrapping on the posts. No slack, no loops: a loose loop is a strangulation risk under 3.	5. The bridge span is 54", not 42" BRIEF span ~42" 162" wall - 60" loft - 48" playhouse = 54" of clear air between them. Both structures are in corners, so neither can slide toward the other to close the gap. Order the net for 54". Expect 6-8" of sag at that span, which is why the plan set carries a top rope AND a mid rope.
3. A 36-38" hobbit door will not fit under a 44" platform BRIEF R.O. 24-26" W x 36-38" H 44" deck - 3/4" plywood - 5-1/2" joist = 37-3/4" of wall, floor to underside of framing. Take out the 1-1/2" bottom plate, a 1-1/2" flat header, a 2-3/4" cripple and a 1-1/2" top plate and 32" is what is left. A 32" door is still right for a 3-6 year old. The semicircular arch on top reads taller than it measures.	THE THREE PHASES 18 mo - 3 yr 3 yr + 5 yr + (or when the youngest is 4) Ground den + playhouse. Ladder stored OFF the structure. Nothing to fall from. Both spaces are at floor level. Hang the ladder. Loft in use. Gate self-closes. Only once he climbs it unassisted, both ways, every time. Unscrew the panel, bolt the eye bolts, hang the net. Two hours. Every provision is already in the wall.

SHEET INDEX A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 Cover · critical notes · phases Overall room plan · anchor zones · studs Loft deck framing · hexagon geometry Loft developed elevation · guards · gate Knee wall · hobbit door · arch template Removable ladder · hook + pin · den shelf Playhouse plan · elevation · roof framing Bridge — DEFERRED. Provisions + load check Cut list Hardware · Home Depot buy list Reference model views Technical orthographic views	AT A GLANCE Deck Angles Posts Guards Toe board Ladder Gate Bridge Lumber Plywood Cost Build 22.36 sq ft hexagonal, 60" x 60" envelope 90/90/157.5/135/157.5/90 — miters 11.25 & 22.5 6 at the loft, 4 at the playhouse 2x2 balusters, 2.59"-2.70" clear, 36" tall 2x4 on edge — no gap at deck level 5 rungs @ 7-5/16" — lifts off, 2 pins self-closing, swings in, no lock deferred — provisions in the wall now ~65 lf 4x4 · ~137 lf 2x4 · ~55 lf 2x2 4 sheets 3/4" \$900 - \$1,150 phase 1 (sheet A9) 2 weekends, one person and a helper	WHAT THIS SET IS MEASURED AGAINST CPSC Public Playground Safety Handbook, Pub. 325 · 3.5" torso probe — no guard opening passes it · 3.5"-9" head entrapment window avoided throughout · toddler platform cap of 32" — why the loft is 3 yr+ · protrusion limit — cap nuts on every through-bolt ASTM F1148, home playground equipment 16 CFR 1213 bunk-bed guardrail gap rule, 3.5" NOT ENGINEER-STAMPED AND NOT CERTIFIED. Designed to these rules, tested against none of them. Verify stud spacing on site before drilling a ledger. Nothing here replaces watching a two-year-old.
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OVERALL ROOM PLAN

structure locations · wall anchor zones · stud layout

A1



LOAD PATH

The wall ledgers carry the real load.
Posts are secondary / aesthetic.

LOFT: two 2x6 PT ledgers, one on the north wall and one on the west wall, 3/8" x 4" lag + washer into EVERY stud.

PLAYHOUSE: north and east posts lag-bolted directly to wall studs.

BRIDGE: both ends through-bolted into 2x6 ledgers that are themselves lagged into a minimum of 2 studs.
Eye BOLTS with washer + nylock nut.
Never screw eyes.

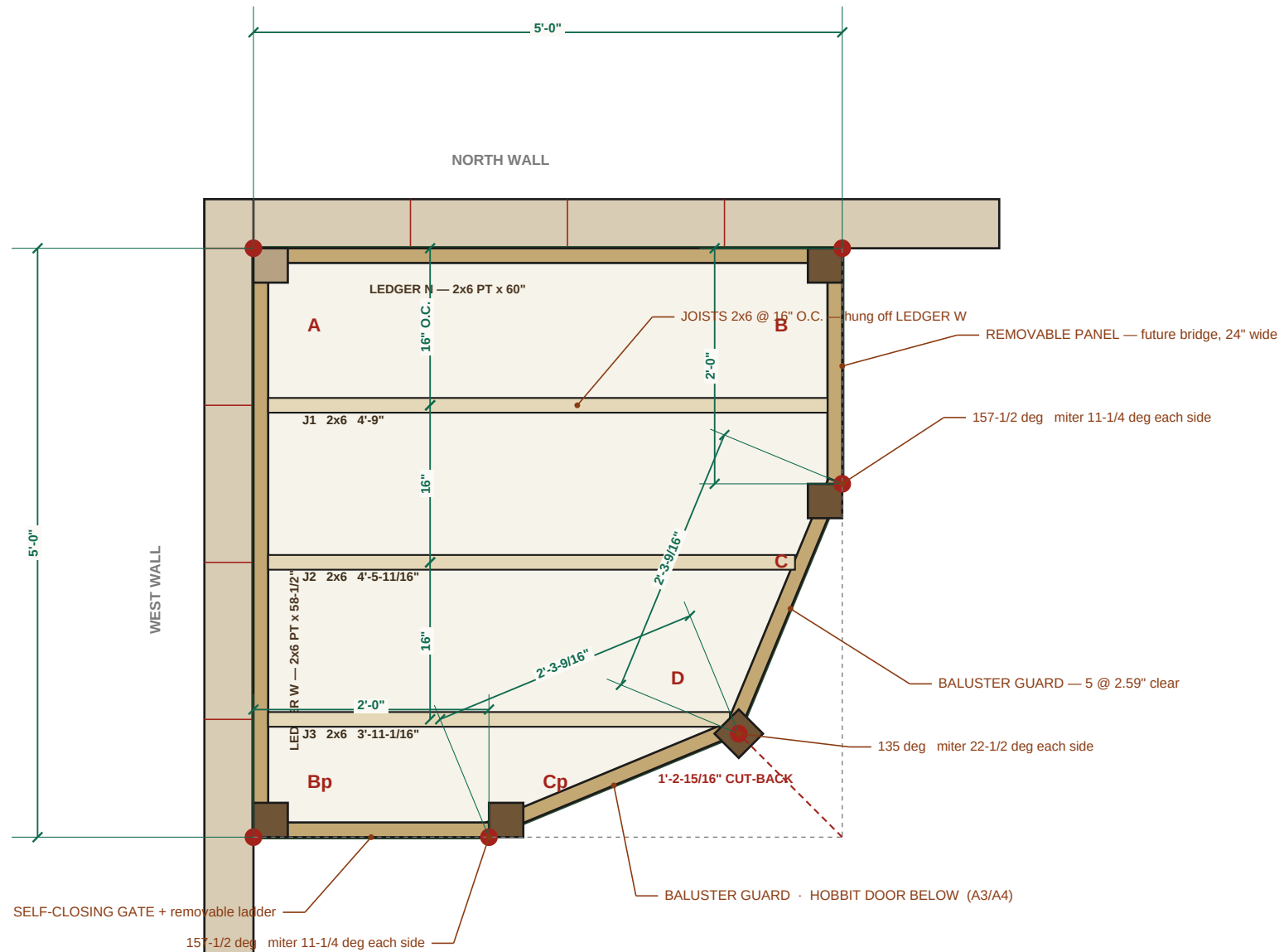
VERIFY STUD SPACING ON SITE.

16" O.C. is assumed throughout; if your studs are 24" O.C. or irregular, shift ledger bolt locations to land on wood and add a bolt — do not rely on drywall anchors anywhere in this set.

LOFT — DECK FRAMING PLAN

hexagon geometry · joist layout · post schedule

A2



HEXAGON GEOMETRY — RESOLVED

VERTEX	u (in)	v (in)	POST CUT
A	0.000	0.000	4x4 x 37-3/4"
B	60.000	0.000	4x4 x 80"
C	60.000	24.000	4x4 x 80"
D	49.456	49.456	4x4 x 80"
Cp	24.000	60.000	4x4 x 80"
Bp	0.000	60.000	4x4 x 80"

WHY THIS SHAPE

A 45-degree chamfer of one corner gives a PENTAGON, not a hexagon — the two cut edges come out collinear, so the middle vertex is not a corner at all.

A true 6-sided deck inside a 60 x 60 corner needs two 22-1/2 degree joints flanking one 45 degree joint. Interior angles: 90 / 90 / 157-1/2 / 135 / 157-1/2 / 90. They sum to 720, as a hexagon must.

Every miter is therefore 11-1/4 or 22-1/2 degrees. Both are detents on any 10-inch miter saw. Nothing in this deck needs a custom angle or a bevel gauge.

DECK AREA 21.75 SQ FT

Both 60-inch wall runs stay full length, so both ledgers get maximum stud engagement. That is the whole point of cutting the outer corner instead of the wall corners.

JOIST NOTES

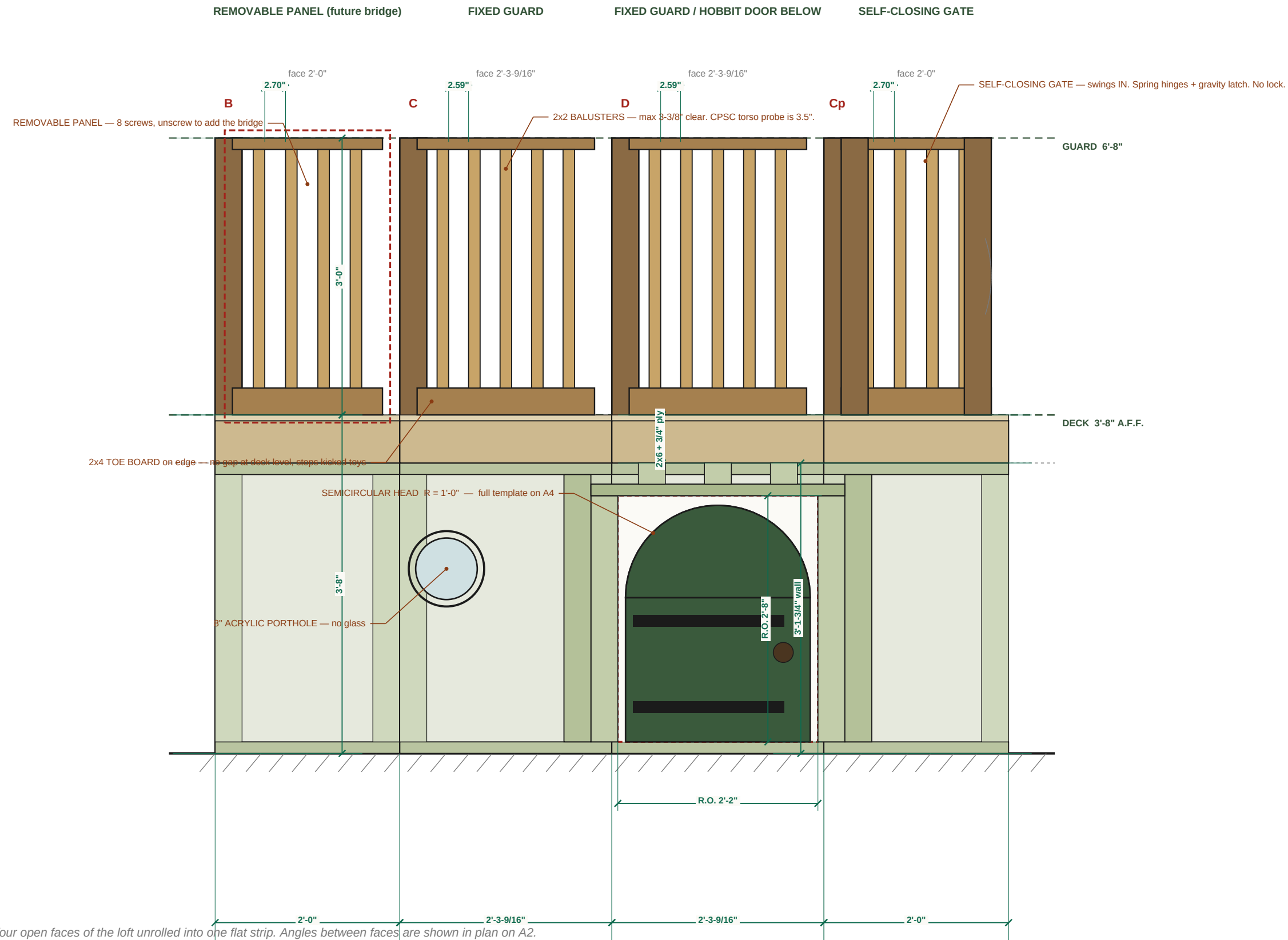
J1 is square both ends — Simpson LUS26 hangers. J2 and J3 die into an angled rim. Use Simpson LSU26 adjustable skewable U hangers (Internet # 100375129), field-bent to the angle shown. Alternatively bevel the joist end and add a 2x6 cleat underneath. Do not toe-nail either one.

POST D sits at 45 degrees so its faces meet both angled rims squarely. Cut its notches before standing.

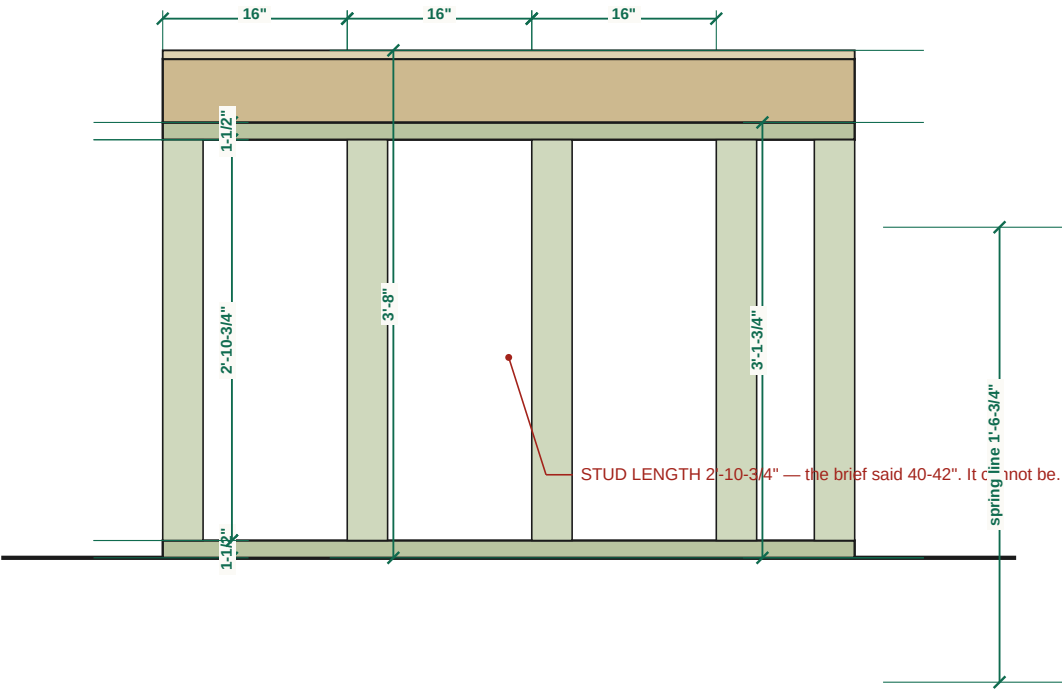
LOFT — DEVELOPED ELEVATION

all four open faces unrolled flat

A3

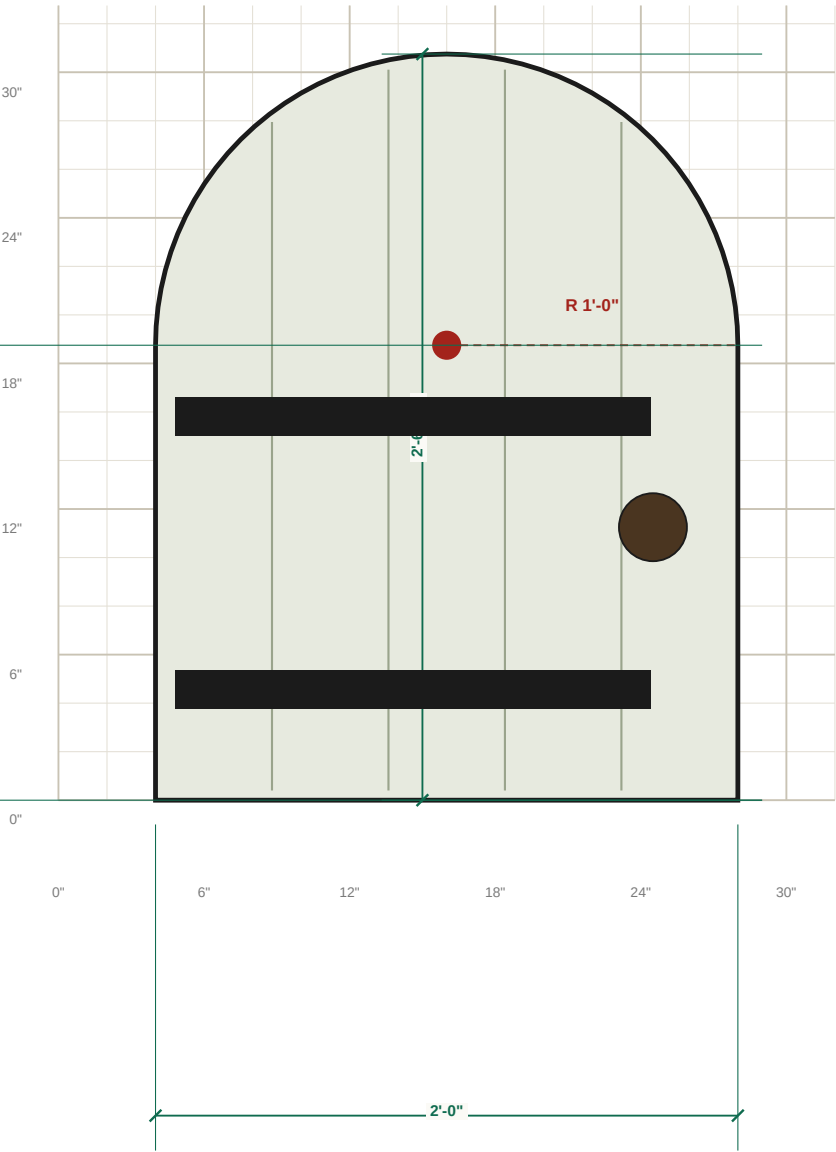


KNEE WALL SECTION — typical



HOBBIT DOOR — FULL TEMPLATE

2" grid — scale off the grid, not the paper



DOOR BUILD-UP

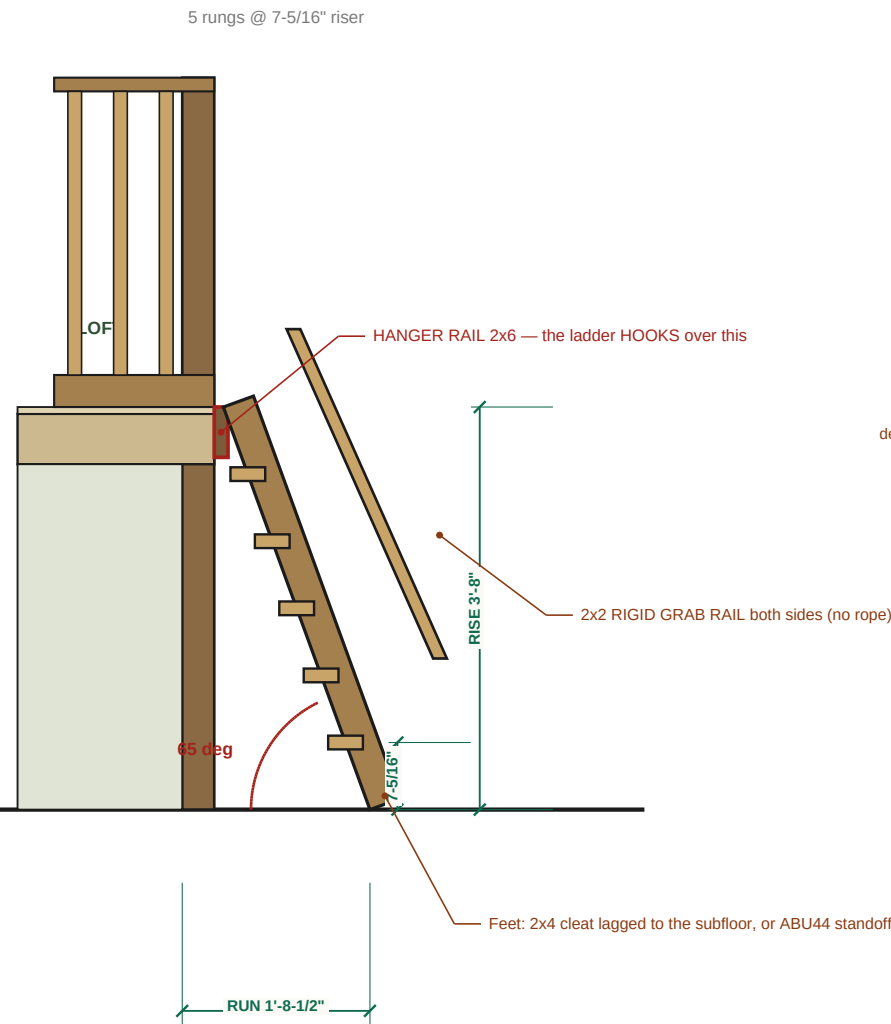
- Slab: two layers of 3/4" ply, glued and screwed, grain crossed, cut to the template above = 1-1/2" thick.
- Face with 1x4 tongue-and-groove pine, vertical, ripped to the arch with a jigsaw.
- Hinges: 12" black strap hinges, 2 per door, through-bolted with 1/4" carriage bolts — screws alone will sag.
- Jamb: 3/4" ply, 3 sides, glued and screwed into the king/jack studs. No sill — remove the bottom plate inside the R.O. after the wall is standing.
- Knob: 2-3/4" round wood knob, 1/4" lag from the back.
- Magnetic catch, not a latch. A kid must be able to push the door open from the inside with no hardware.
- NO lock of any kind, inside or out.
- Paint: deep green (see A9). Two coats, sand between.

REMOVABLE LADDER + DEN SHELF

elevation · hook and pin detail · ground-level shelf

A5

LADDER ELEVATION — LIFTS OFF

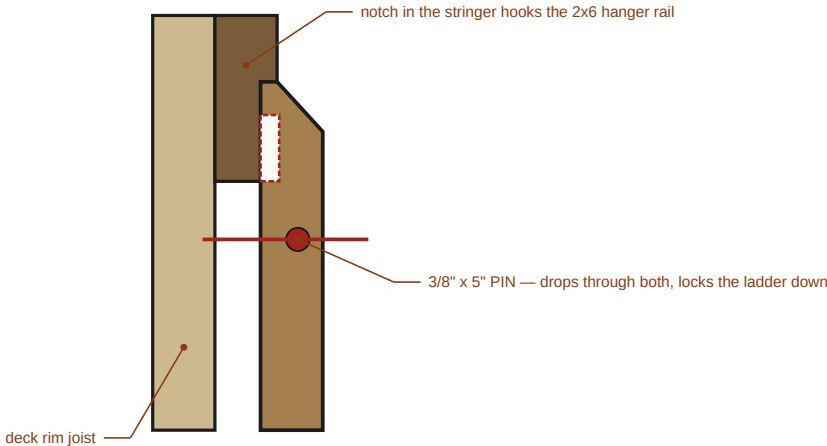


WHY THE LADDER LIFTS OFF

A gate stops a fall. It does not stop an 18-month-old, because whatever latch a 4-year-old can work, a determined toddler eventually works too.

The ladder is the real access control. Pull two pins, lift it off, stand it behind the playhouse. The loft is then physically unreachable — there is nothing to climb.

Hang it when the 4-year-old is up there. Pull it at night and any time the little one is loose in the room.

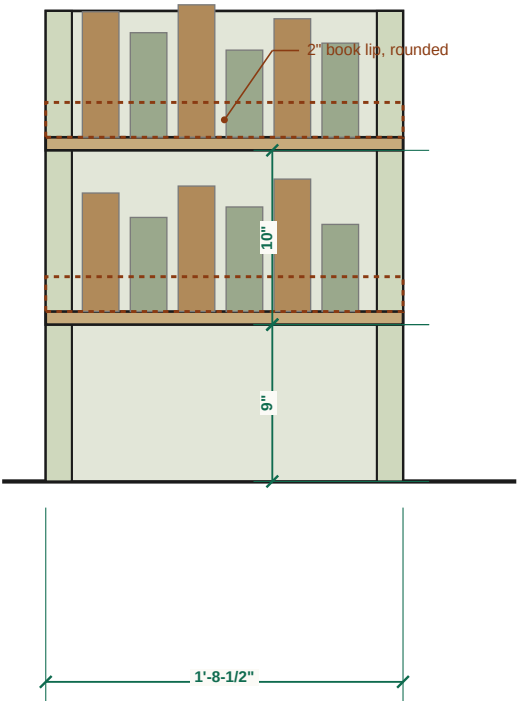


HOOK + PIN DETAIL

pull two pins, lift the ladder off, store it flat

DEN SHELF — GROUND LEVEL

inside the enclosed base, NOT in the guardrail



WHY THE SHELVES CAME OUT OF THE RAILING

You were right, and for a better reason than you think. A shelf bolted into a guardrail is a ladder. A 4-year-old puts a foot on the bottom shelf and their centre of mass clears a 36" rail. The guard stops working.

The books and the apothecary jars move to floor level, inside the den, where the 18-month-old actually plays.

CHOKING — READ THIS

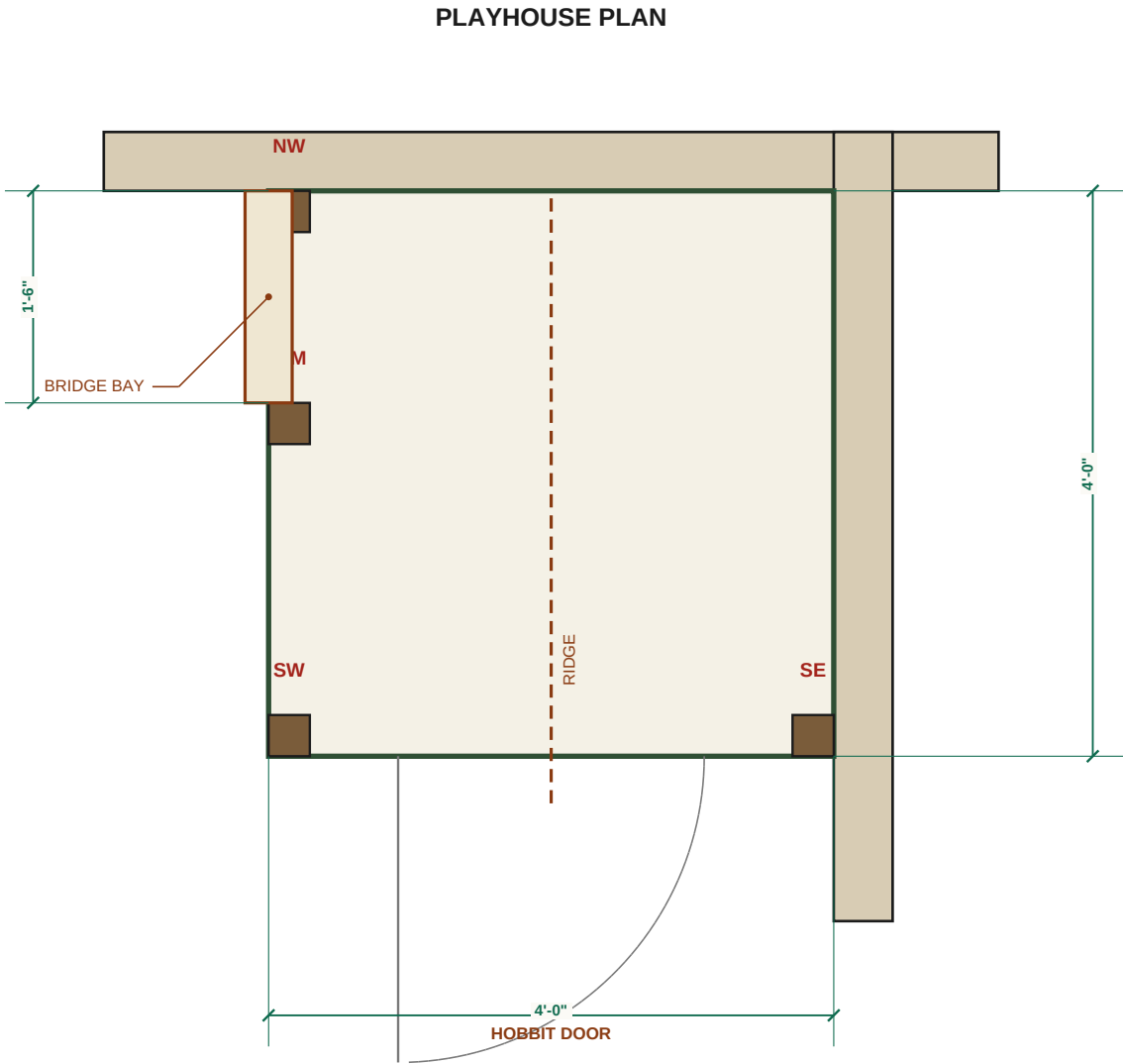
Acorns, dried botanicals and anything under 1-3/4" is a choking hazard until 3. Either hot-glue the jar lids shut so they are display-only, or leave the jars out until the little one is past it. Board books only down here.

PLAYHOUSE

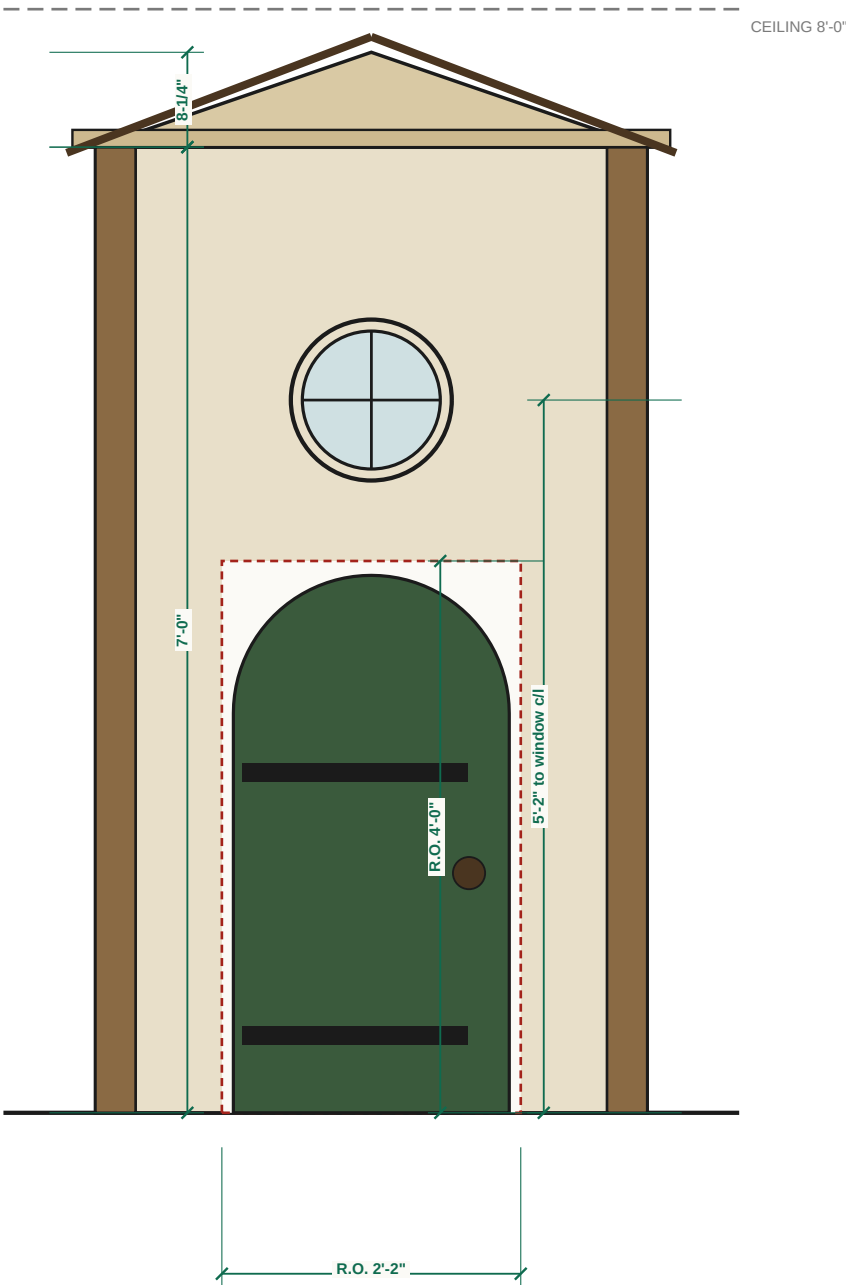
plan · south elevation · roof framing

A6

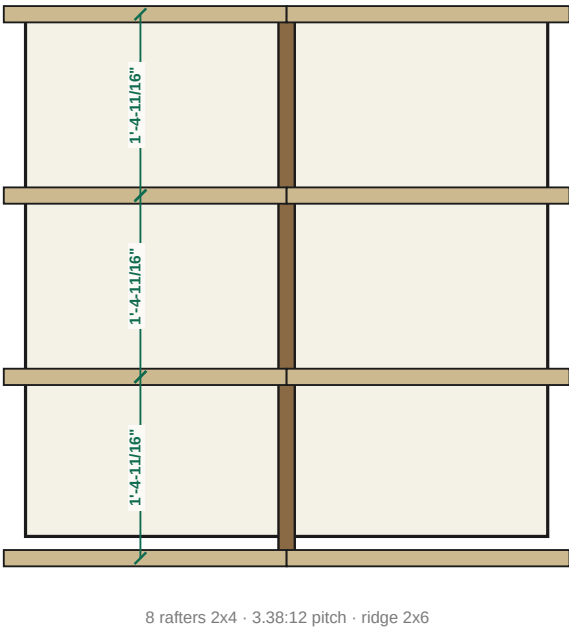
PLAYHOUSE PLAN



SOUTH ELEVATION (faces the room)



ROOF FRAMING PLAN



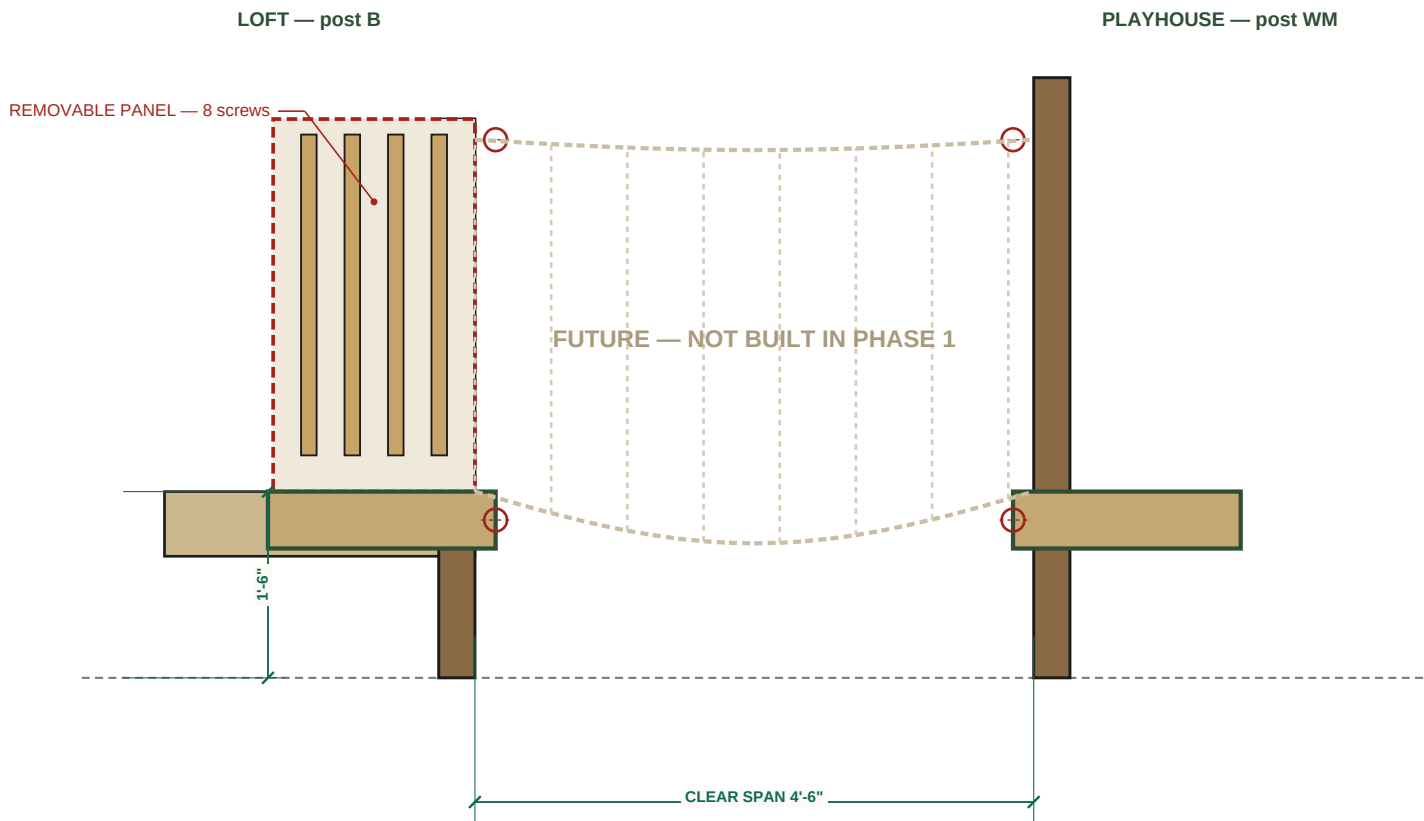
PLAYHOUSE NOTES

WALLS STOP AT 84", NOT 96".

A 96"-tall box under a 96" ceiling has no room for a roof. Dropping the wall to 84" buys a real 10-1/2" peaked roof that reads from across the room, and still leaves 84" of interior headroom.

- NW and SE posts lag directly to wall studs: 3x 3/8" x 4" lag + washer each, staggered.
- SW post is free-standing — anchor with a Simpson ABU44Z post base lagged to the subfloor.
- WM post creates the 18" bridge bay and matches the loft gate width. Do not omit it.
- Rope-wrap the SW post full height: 1/2" poly rope, spiral-wound, hot-glued every 6".
- Rough-saw look: run a wire wheel over the 4x4s, then a coat of dark walnut stain wiped back.
- Interior: battery LED lantern on a hook, no wiring.
- Leave the west wall below the bridge bay open or slatted so the space does not feel like a closet.

BRIDGE — FUTURE PROVISIONS



INSTALL THESE THREE THINGS NOW. THEY COST ABOUT \$40 AND SAVE A TEARDOWN LATER.

1. Both 2x6 PT wall ledgers, lagged into studs. They read as a trim band until the day you use them.
2. The playhouse WM post at 18" off the north wall. It makes the bay. Skip it and you are cutting into finished siding later.
3. The 2x6 net headers as blocking behind the loft rim and the playhouse west face.

ADDING THE BRIDGE LATER IS THEN A TWO-HOUR JOB: unscrew the 8-screw panel, drill 8 holes, bolt 8 eye bolts, hang the net.

I CHECKED THE BRIDGE BEFORE DELETING IT

You asked whether it holds a 4-year-old. It does, easily.

Two 42 lb kids, x5 dynamic factor for bouncing = 420 lb, split across 4 anchors = 105 lb per anchor.

A 3/8" FORGED shoulder eye bolt is good for ~1000 lb in line. Eight times the load.

The 3/8" lags into studs give ~587 lb each in withdrawal; four per ledger is about eleven times what is needed.

Strength was never the problem. Two other things were.

WIDTH

18" is a balance beam, not a bridge. Playground crossings run 20-24". So the loft gate face went from 20" to 24", which also made the deck slightly BIGGER, 22.36 sq ft. The bridge you add later will be 24" wide.

MESH SIZE — THE ONE THAT MATTERS

CPSC treats any opening between 3.5" and 9" as a head entrapment: a small body passes through, the head does not. A 4-6" cargo net mesh fails both probes. That is fine for a supervised 6-year-old and wrong for a 2-year-old in the same room. When you buy the net, get mesh under 3.5", or add a fine secondary net on both faces.

WAIT UNTIL THE YOUNGEST IS 4. The bridge is the single most dangerous element in the whole build, and the one you lose the least by deferring.

CUT LIST

every member, every finished length

A8

CUT LIST — generated directly from the model. Every length is the finished cut length.

Lengths are long-point to long-point where a member is mitered. Cut one of any repeated part, test-fit it, then batch the rest.

LOFT / POSTS

STOCK	LENGTH	QTY	NOTE
4x4 PT	3'-1-3/4"	1	room-corner stub
4x4 PT	6'-8"	1	bridge post
4x4 PT	6'-8"	3	
4x4 PT	6'-8"	1	ladder post

LOFT / LEDGERS

STOCK	LENGTH	QTY	NOTE
2x6 PT	5'-0"	1	lag to studs @16 O.C., 3/8 x 4 lag + washer
2x6 PT	4'-10-1/2"	1	lag to studs @16 O.C.

LOFT / RIM

STOCK	LENGTH	QTY	NOTE
2x6	2'-0"	2	outside face length 24.000"
2x6	2'-3-9/16"	2	outside face length 27.553"

LOFT / JOISTS

STOCK	LENGTH	QTY	NOTE
2x6	4'-9"	1	LUS26 hanger both ends
2x6	4'-5-11/16"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	3'-11-1/16"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	1'-5"	1	decking support at 135-deg corner

LOFT / KNEE WALL

STOCK	LENGTH	QTY	NOTE
2x4 PT	2'-0"	2	24.00" face length
2x4	2'-0"	2	24.00" face length
2x4	2'-10-3/4"	8	34.75" - NOT 40-42"
2x4 PT	2'-3-9/16"	2	27.55" face length
2x4	2'-3-9/16"	2	27.55" face length

LOFT / HOBBIT DOOR

STOCK	LENGTH	QTY	NOTE
2x4	2'-8"	2	32.00" jack
2x4	2'-10-3/4"	2	34.75" king
2x4	2'-9"	1	flat - deck load bypasses this wall
2x4	2-3/4"	3	2.75"

LADDER

STOCK	LENGTH	QTY	NOTE
2x6	1'-6"	1	lag to rim + both posts; ladder hooks over this, 3/8
2x4	4'-0-9/16"	2	48.55" o/a; notch top to hook the hanger rail, 25 de
2x2	3'-5-7/16"	2	1/2" roundover, sand 180. RIGID - sagging rope is a
2x2	3-3/4"	4	grab-rail standoff block
2x4	1'-3"	5	riser 7.33"; dado 3/4" into stringers + 2 screws/end

LOFT / GUARDS

STOCK	LENGTH	QTY	NOTE
2x4	1'-7-1/2"	1	REMOVABLE - 8 screws. Take it off to add the bridge
2x4	1'-7-1/2"	2	on edge at deck level; closes the bottom gap, stops
2x2	2'-7"	8	2.700" clear gap - CPSC limit is 3.5"; 1/4" roundove
2x4	1'-11-1/16"	2	fixed guard; top of rail = 80" = 36" above deck
2x4	1'-11-1/16"	2	on edge at deck level; closes the bottom gap, stops
2x2	2'-7"	10	2.592" clear gap - CPSC limit is 3.5"; 1/4" roundove
2x4	1'-7-1/2"	1	SELF-CLOSING GATE leaf - hinged at Cp, swings IN ove
2x4	3'-0"	2	self-closing spring hinges (pr) / gravity latch + ma

LOFT / DEN SHELF

STOCK	LENGTH	QTY	NOTE
3/4" ply	1'-5-1/2"	2	17.5" x 7.25"; screw to studs, 1/4" roundover, 2" li
1x3	1'-5-1/2"	2	book lip, rounded

PLAYHOUSE / POSTS

STOCK	LENGTH	QTY	NOTE
4x4 PT	7'-0"	1	exposed room corner
4x4 PT	7'-0"	1	bridge bay post
4x4 PT	7'-0"	1	lag to N wall studs
4x4 PT	7'-0"	1	lag to E wall studs

PLAYHOUSE / PLATES

STOCK	LENGTH	QTY	NOTE
2x4	4'-0"	4	top plate @ 84" - anchors roof + trim

PLAYHOUSE / ROOF

STOCK	LENGTH	QTY	NOTE
2x6	4'-2"	1	1x6 or 2x6 ridge, 1" below ceiling
2x4	2'-3"	8	3.38:12 pitch, 2" overhang

PLAYHOUSE / DOOR

STOCK	LENGTH	QTY	NOTE
2x4	4'-0"	2	48" jack
2x4	7'-0"	2	84" king
2x6	2'-9"	1	double 2x6 header over 26" R.O.

BRIDGE / FUTURE PROVISIONS

STOCK	LENGTH	QTY	NOTE
2x6 PT	1'-9-1/2"	2	INSTALL NOW. Lag 3/8x4 into 2 studs min. Eye bolts g
2x6	2'-0"	2	INSTALL NOW as blocking. Drill for 4x 3/8 eye bolts

SHEET GOODS + WHAT TO ACTUALLY BUY

ITEM	NET	BUY
4x4 x 8 ft PT post	64.5 lf	8
4x4 x 10 ft PT post	(1 post + the stub)	1
2x6 x 8 ft Ground Contact PT	13.5 lf	3
2x6 x 8 ft SPF	35.6 lf	5
2x4 x 8 ft Ground Contact PT	8.6 lf	2
2x4 x 8 ft SPF stud	136.9 lf	18
1x3 x 8 ft pine (book lips)	2.9 lf	1
3/4" 4x8 ply — deck + shelf	21.75 sq ft + shelf	2
3/4" 4x8 ext ply — skins + doors	~46 sq ft	2
1x4 T&G pine — door faces	~24 lf	1 bdl

2x quantities include 15% waste. Every cut in this set fits an 8 ft stick.
The one 10 ft 4x4 yields an 80" post AND the 37-3/4" corner stub with almost no waste. Nine 8-footers would leave you one post short.

HARDWARE SCHEDULE + HOME DEPOT BUY LIST

Internet # verified against homedepot.com. Prices are US national ranges and move weekly — treat them as a budget, not a quote. Items without an Internet # are commodity stock carried in every store; grab them off the shelf.

ITEM	INTERNET #	QTY	EST. EACH	WHAT IT DOES / WHY THIS ONE
PHASE 1 LUMBER				
4 in. x 4 in. x 8 ft. #1 Pressure-Treated Post	202812131	8	\$12 - \$16	four loft posts at 80" + four playhouse posts at 84"
4 in. x 4 in. x 10 ft. #1 Pressure-Treated Post	—	1	\$16 - \$22	fifth 80" post AND the 37-3/4" corner stub, almost no waste
2 in. x 6 in. x 8 ft. Ground Contact PT	—	3	\$9 - \$13	loft ledgers + BOTH future-bridge ledgers (install now)
2 in. x 6 in. x 8 ft. Prime Kiln-Dried SPF	—	6	\$7 - \$10	joists, rim, ladder hanger rail, playhouse header, ridge
2 in. x 4 in. x 8 ft. Ground Contact PT	—	2	\$6 - \$9	knee wall bottom plates (they sit on the subfloor)
2 in. x 4 in. x 8 ft. Prime Kiln-Dried Stud	—	20	\$4 - \$6	studs, plates, rafters, ladder, top rails, toe boards
2 in. x 2 in. x 8 ft. Furring / S4S Pine	—	9	\$3 - \$5	GUARD BALUSTERS + ladder grab rails — ~55 lf
1 in. x 3 in. x 8 ft. Pine Board	—	1	\$6 - \$9	den shelf book lips
3/4 in. 4x8 Sanded Plywood (BC or Sande)	—	4	\$48 - \$68	2 structural, 2 skin + doors + den shelf
1 in. x 4 in. Tongue & Groove Pine (bundle)	—	1	\$28 - \$40	door faces, both doors
STRUCTURAL HARDWARE				
Simpson Strong-Tie LUS26 Face-Mount Joist Hanger	100374889	8	\$1.77 ea	square joist-to-ledger connections
Simpson Strong-Tie LSU26 Adjustable Skewable U Hanger	100375129	6	\$4 - \$6 ea	J2/J3 into the angled rims — THE part for this
Simpson Strong-Tie ABU44Z Adjustable Standoff Post Base	100375358	3	\$14 - \$19 ea	free-standing posts: loft D, playhouse SW, ladder feet
Simpson Strong-Tie A23 / A35 Framing Angle	—	16	\$0.80 ea	post-to-rim and plate ties
Everbilt 3/8 in. x 4 in. Hex Lag Screw (bulk)	—	60	\$0.55 ea	ledgers into studs, posts into studs
Everbilt 3/8 in. Flat Washer (bulk)	—	100	\$0.15 ea	every lag and every bolt, both sides
Everbilt 1/4 in. x 3 in. Hex Lag Screw	—	40	\$0.40 ea	guards to posts, hinges, hanger rail, trim
Everbilt 3/8 in. Acorn / Cap Nut	—	20	\$0.45 ea	CPSC protrusion rule — cap every bolt end a kid can hit
#9 x 3 in. Star Drive Construction Screw, 5 lb box	—	1	\$34 - \$44	general framing
#8 x 1-5/8 in. Construction Screw, 1 lb box	—	2	\$9 - \$12	decking, skins, balusters — SCREWED, never nailed
LADDER + GATE				
Everbilt 3/8 in. x 5 in. Clevis / Hitch Pin w/ lanyard	—	2	\$4 - \$7 ea	lock the ladder to its hanger rail. Pull 2, lift it off.
Everbilt 3 in. Self-Closing Spring Hinge (pair)	—	1	\$12 - \$18	the deck gate MUST close itself. Do not use plain hinges.
Everbilt Gravity / Gate Latch, black	—	1	\$8 - \$14	NO lock, NO keyed hardware, nothing a kid can shut on themselves
Everbilt Magnetic Push Catch	—	3	\$4 - \$6	both hobbit doors + the gate. Kid must push out from inside.
ROPE — DECORATIVE ONLY NOW				
Everbilt 1/2 in. x 50 ft. Twisted Poly Rope (manila look)	—	1	\$22 - \$32	TIGHT post wraps and rail whipping. No slack, no hanging loops.
Hot glue sticks / exterior construction adhesive	—	1	\$6 - \$10	bed every rope wrap so it cannot be unwound into a loop
DOORS, FINISH, LIGHT				
NUVO IRON 12 in. Black Steel Strap Hinges Kit (2-pack)	320202663	2	\$18 - \$26	one 2-pack per hobbit door
2-3/4 in. Round Wood Cabinet Knob	—	2	\$4 - \$7	hobbit door knobs
8 in. Clear Acrylic Circle / Plexiglass sheet to cut	—	1	\$12 - \$22	porthole — acrylic only, never glass
12 in. Clear Acrylic Circle	—	1	\$16 - \$28	playhouse round window
Commercial Electric LED Puck Light, battery, 3-pack	—	2	\$18 - \$26	2 in the den, 1 in the playhouse. Batteries behind a screwed cover.
BEHR Premium Plus Interior Eggshell — deep green	—	1 qt	\$20 - \$26	match "Dark Everglade" / "Pine Mountain"
Varathane Dark Walnut Wood Stain, qt	—	1	\$14 - \$19	rough-sawn look on the 4x4s and trim
Minwax Polycrylic Clear Satin, qt	—	1	\$20 - \$26	water-based — low odor, and it is a kid's room
Zero-VOC primer, qt	—	1	\$16 - \$22	everything a kid touches, and they mouth everything at 18 months
FALL PROTECTION — NOT OPTIONAL				
2 in. Folding Gymnastics / Tumbling Mat, 4 x 8 ft	—	1	\$90 - \$150	A 44" fall needs ~2". 3/4" foam tiles attenuate about 2 ft — NOT enough.
3/4 in. Interlocking Foam Mat Tiles, 24x24, 6-pack	—	2	\$26 - \$38	general floor, and under the playhouse door threshold

KIT	PHASE 3 — BUY ONLY WHEN YOU ADD THE BRIDGE	ns.py / geometry.py / Blender 4.2			NOT ENGINEER-STAMPED · VERIFY STUDS ON SITE · SEE A0
Hardware Essentials	3/8 x 6 in. FORGED Galvanized Eye Bolt (5-pk)	203809978	2	\$18 - \$26	FORGED shoulder pattern. Bent zinc eye bolts are 3x weaker.
Everbilt	3/8 in. Nylon Insert Lock Nut (nylock)	—	12	\$0.35 ea	back side of every eye bolt
Cargo net,	24 in. wide, mesh UNDER 3.5 in.	—	1	\$70 - \$130	mesh 3.5-9 in. is a head-entrapment size. Measure before you buy.

REFERENCE VIEWS — MODEL

rendered from the Blender model, dimensionally accurate

A10

MODEL VIEWS — the geometry, not a mood board. Photoreal 2K versions of all four exist in Higgsfield: exports/higgsfield/MANIFEST.md



01 Isometric overview — both structures and the crossing



02 From the room doorway — what a kid sees walking in



03 Loft three-quarter — hex deck, rope guard, bookshelf



04 Playhouse front — hobbit door, round window, peaked roof

ORTHOGRAPHIC + FRAMING-ONLY VIEWS



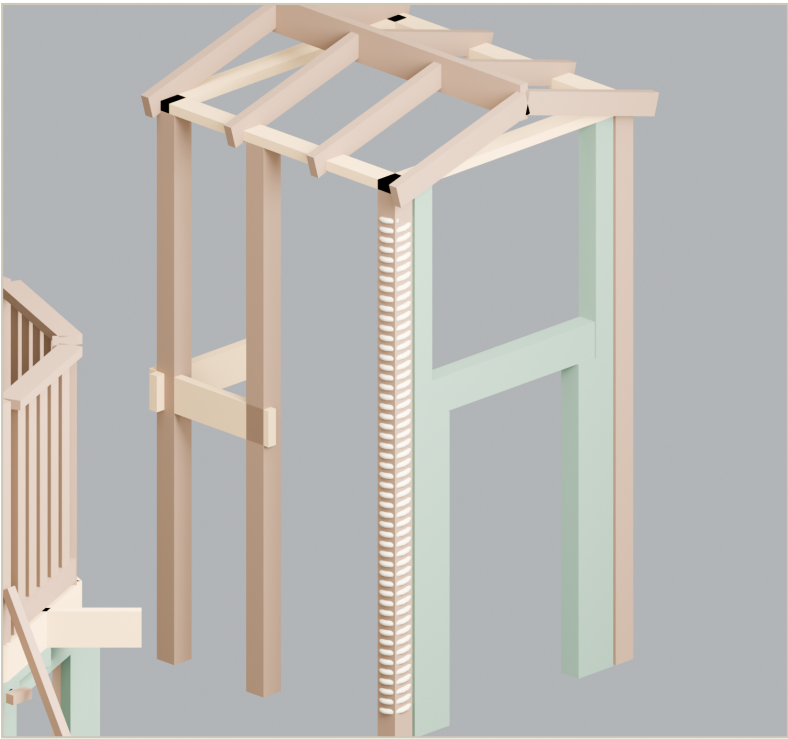
T1 True plan — walls and ceiling removed, floor retained



T2 North elevation — the wall both structures anchor to



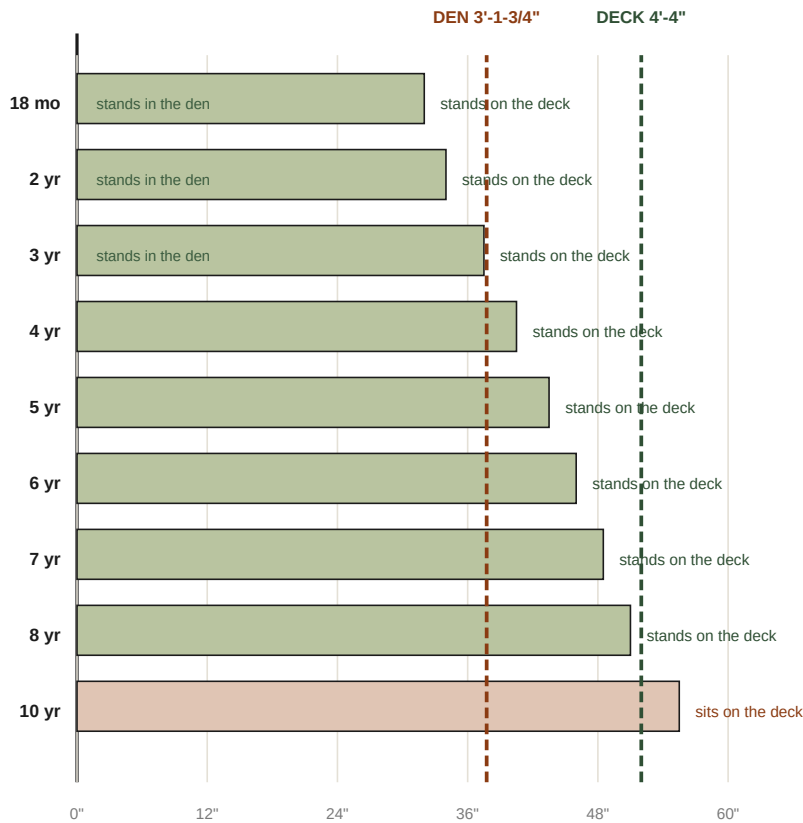
T3 Loft framing only — skins, decking and all rope hidden



T4 Playhouse framing only — posts, plates, rafters, door bucks

WILL THIS STILL BE FUN IN FIVE YEARS? — fit by age, and the things that only show up after you live with it

Heights are CDC 50th percentile standing height. Your kid will not be average; add or subtract and re-read the table.



STANDING HEADROOM vs AGE

WHAT THE CHART SAYS

- The DEN is a stand-up room only until about 3. After that it is a sit-down cave, which is exactly what a 5-year-old wants it to be. It never stops working; it just changes job.
- The DECK takes a standing child to about 8. Past that they sit up there, which is still where kids read and hide at that age.
- An adult can never stand on the deck. That is deliberate. If you could, they would want you up there, and 22 sq ft is a two-kid deck.

THE HONEST LIFESPAN: 18 MONTHS TO ROUGHLY 9 OR 10.

The bridge in phase 3 is what buys the last three years of it.

ADULT ACCESS — THE ONE PEOPLE FORGET

If a two-year-old bumps their head inside the den, you have to get to them. The hobbit door is 24-1/2" x 31-1/4" finished. An adult CAN crawl it, barely. So: screw the B-C knee-wall skin panel on, never glue and never nail it. Eight screws and the whole panel comes off in about a minute. Mark the screw heads with a dab of paint so you can find them in a hurry.

AIR

Plywood floor above, plywood skin around, one sealed acrylic porthole. Drill a pattern of 1-1/4" holes near the top of two panels. Read as a design detail, and they vent the box. Use 1-1/4", never 3/8" to 1" — that range is a finger-entrapment size.

SPLINTERS BEAT AESTHETICS AT THIS AGE

Rough-sawn is the look you want and the worst possible surface for an 18-month-old who mouths everything. Wire-brush and stain only what nobody grips: posts ABOVE the deck, roof trim, the gable. Sand every graspable surface to 150-180 — balusters, top rails, rungs, grab rails, door edges, the whole den interior. Run a bare forearm over every surface below 40" before you let him in.

STORING THE LADDER

Lay it FLAT behind the playhouse or strap it to a wall. A ladder leaning against a wall is the single most tippable object in the room, and a toddler will pull on it precisely because it is new and out of place.

NO CORDS. NONE.

Battery puck lights only, with the battery cover screwed shut. No plug-in string lights, no extension cords, no lamp on the deck. A cord at 44" is a strangulation hazard and a trip hazard at once.

RE-CHECK SCHEDULE — write the date on the ledger in pencil

30 days: re-torque every lag. Green lumber shrinks and they WILL be loose. Every 6 months: lags again, gate closes and latches, ladder pins present, no baluster gap grown past 3-1/2", no splinters, no protruding screws. Every spring: re-coat anything that has worn through to bare wood.